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Palladium-Catalyzed Formylation of Alkenylzinc Reagents with *S*-(4-Nitrophenyl) Thioformate

Ryosuke Haraguchi,* Sho-go Tanazawa, Naoya Tokunaga, and Shin-ichi Fukuzawa*

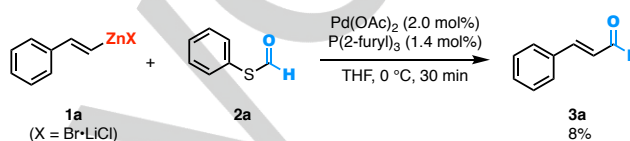
Abstract: We report the synthesis of enal compounds by palladium-catalyzed formylation of alkenylzinc reagents with *S*-(4-nitrophenyl) thioformate. Various functional groups were tolerated in the present reaction. ¹H NMR experiments revealed that the products had a non-protected formyl group, which can be utilized for further C–C bond formation reactions. We successfully achieved the one-pot synthesis of a 1,5-diene-3-ol compound via sequential formylation and allylation.

Introduction

α,β -Unsaturated aldehydes are useful intermediates in organic synthesis as they can be employed as Michael acceptors or reactive carbonyl groups.¹ Among the number of synthetic routes to these compounds,^{2–4} introduction of a formyl group into organic molecules is considered as the most straightforward approach.^{5–9} Formylation of organolithium or organomagnesium reagents with formamide derivatives is the simplest protocol for the preparation of enal derivatives, although their high nucleophilicity makes it difficult to synthesize the functionalized enal compounds.⁵ Alternatively, the transition-metal-catalyzed hydroformylation of alkynes has been achieved by several research groups.⁶ Moreover, palladium- and nickel-catalyzed reductive formylation of vinyl halides or pseudohalides has been also reported.⁷ However, harsh reaction conditions and the use of uncommon ligands and additives are the practical limitations of these synthetic methods. Moreover, the functional tolerability of these reactions in most cases has not been studied well. Thus, the development of synthetic methods for the preparation of functionalized enals is still sought after. Recently, Mariano, Xie, and Wang reported the organocatalytic formylation of aryl- and alkenylboronic acids, which has excellent functional tolerability.⁹

We have recently developed the palladium-catalyzed formylation of arylzinc reagents with *S*-phenyl thioformate (Fukuyama-type formylation).^{10,11} As a logical extension of this report, we focused on the palladium-catalyzed formylation of alkenylzinc reagents. However, we expected this formylation to be much more difficult than that of the arylzinc reagents because of a side reaction of the enal products with a thiolate anion generated *in situ*. As expected, the reaction of alkenylzinc reagent **1a** with thioformate **2a** under the optimized reaction conditions employed in the formylation of arylzinc reagents yielded only 8% of the product (Scheme 1), with the formation of the Michael adduct. Thus, the choice of formylating reagents and optimization of the catalytic reaction conditions are crucial for the success of this reaction. Herein, we report the palladium-catalyzed formylation of alkenylzinc reagents with *S*-(4-

nitrophenyl) thioformate, which dramatically promoted the present reaction. Under mild reaction conditions, the formylation reaction was completed within 10 min to afford the enal derivatives in high yields. ¹H NMR monitoring of the reaction mixture revealed that a formyl group of the enal product existed in a non-protected form. We also achieved the one-pot synthesis of the 1,5-diene-3-ol compound via the formation of two C–C bonds by sequential formylation and allylation.



Scheme 1. Formylation of alkenylzinc reagent **1a** with thioformate **2a**.

Results and Discussion

Initially, we screened the formyl sources, ligands, solvents, and additives used in the palladium-catalyzed formylation of alkenylzinc reagent **1a**. We found that the treatment of **1a** with *S*-(4-nitrophenyl) thioformate (**2b**) in the presence of Pd(OAc)₂ (5.0 mol%), P(3,5-F₂C₆H₃)₃ (9.0 mol%), and ZnBr₂ (1.0 eq) in THF at –20 °C for 10 min gave enal **3a** in 87% yield. The important reaction parameters for the present reaction are shown in Table 1. Tri(2-furyl)phosphine, which is the most effective ligand for the palladium-catalyzed formylation of arylzinc reagents,¹¹ decreased the chemical yield (entry 2). Although PPh₃ gave the desired product in high yield (entry 3), other monodentate phosphine ligands bearing electron-donating or electron-withdrawing substituents provided the desired product in somewhat lower yields (entries 4–8). Bidentate phosphine ligands such as dppf and binap were not effective for the present reaction (entries 9 and 10). The use of zinc bromide as an additive is the most effective although other zinc salts also increased the reaction efficiency to some extent (entries 1 and 11–14).¹² Notably, the efficiency of this reaction relied on the choice of thioformates. We examined various thioformates **2a–f** in the formylation reaction, and found that the thioformates bearing electron-withdrawing substituents gave **3a** in good yield (entries 1 and 16), while those with electron-rich aromatic rings or an alkyl substituent decreased the chemical yields (entries 17–19).

We next investigated the reaction profile using different thioformates **2a–d** in the present formylation. As shown in Figure 1, the reaction rate was dramatically enhanced when **2b** was used as the formylating reagent.

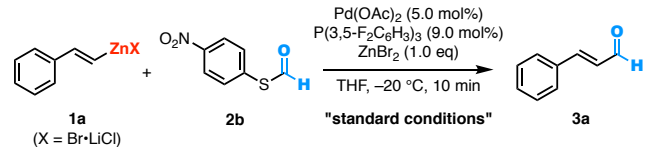
The scope of the alkenylzinc reagents was examined (Table 2). Alkenylzinc reagents bearing diverse functional groups on the arene ring underwent the palladium-catalyzed formylation to afford the corresponding products in high yield. Halogen groups (entries 2 and 3) and an acetal group (entry 8) remained intact under these reaction conditions. The present reaction also

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tolerated a benzyloxy group (entry 9), which may decompose during the reductive palladium-catalyzed formylation using H_2 and CO to give a phenol compound. The highly coordinating amino group did not inhibit the palladium-catalysis (entry 7). Substrates bearing heteroaromatic moieties also gave the corresponding enal compounds in good yields (entries 10 and 11).

Table 1. Effect of reaction parameters on the formylation of alkenylzinc reagent **1a** with **2a**.^[a]

		
Entry	Variation from the standard conditions	Yield ^[b]
1	None	87
2	P(2-furyl) ₃ , instead of P(3,5-F ₂ C ₆ H ₃) ₃	65
3	PPh ₃ , instead of P(3,5-F ₂ C ₆ H ₃) ₃	81
4	P(4-OMeC ₆ H ₄) ₃ , instead of P(3,5-F ₂ C ₆ H ₃) ₃	33
5	P(4-CF ₃ C ₆ H ₄) ₃ , instead of P(3,5-F ₂ C ₆ H ₃) ₃	64
6	P(2-MeC ₆ H ₄) ₃ , instead of P(3,5-F ₂ C ₆ H ₃) ₃	56
7	P(3,5-MeC ₆ H ₃) ₃ , instead of P(3,5-F ₂ C ₆ H ₃) ₃	59
8	PCy ₃ , instead of P(3,5-F ₂ C ₆ H ₃) ₃	51
9	dppf, instead of P(3,5-F ₂ C ₆ H ₃) ₃	58
10	Binap, instead of P(3,5-F ₂ C ₆ H ₃) ₃	57
11	ZnCl ₂ , instead of ZnBr ₂	67
12	ZnI ₂ , instead of ZnBr ₂	73
13	Zn(OAc) ₂ , instead of ZnBr ₂	70
14	no ZnBr ₂	44
15	2a , instead of 2b	22
16	2c , instead of 2b	60
17	2d , instead of 2b	7
18	2e , instead of 2b	5
19	2f , instead of 2b	4

[a] Standard conditions: **1a** (0.24 mmol), **2b** (0.20 mmol), Pd(OAc)₂ (10 μ mol), P(3,5-F₂C₆H₃)₃ (18 μ mol), ZnBr₂ (0.20 mmol) in THF at -20 °C for 10 min. [b] Determined by ¹H NMR analysis of the crude reaction mixture using dibromomethane as the internal standard.

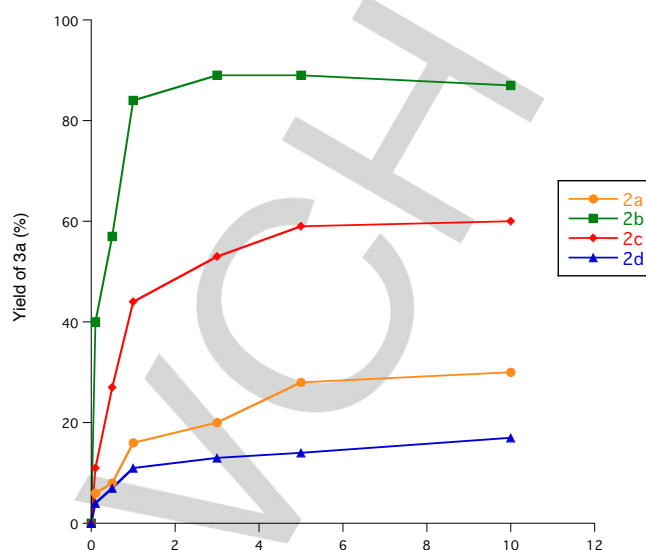
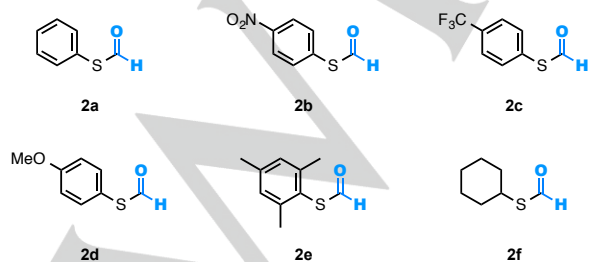
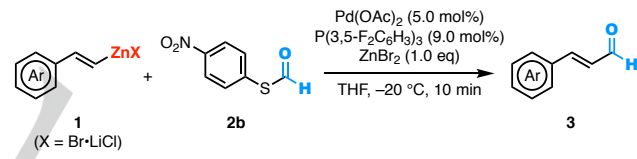


Figure 1. NMR Yields versus reaction time profile using various thioformates **2a-d**.

Table 2. Substrate Scope of Alkenylzinc Reagents.^[a]

			
Entry	Ar	Product	Yield ^[b]
1	C ₆ H ₅	3a	87
2	<i>p</i> -FC ₆ H ₄	3b	80
3	<i>p</i> -ClC ₆ H ₄	3c	93
4	<i>p</i> -MeC ₆ H ₄	3d	95
5	<i>o</i> -MeC ₆ H ₄	3e	98
6	2-naphthyl	3f	92
7	<i>p</i> -NMe ₂ C ₆ H ₄	3g	59
8	3,4-dioxycanC ₆ H ₄	3h	94
9	<i>p</i> -OBnC ₆ H ₄	3i	77
10	2-thienyl	3j	92
11	9-Et-3-carbazolyl	3k	74

[a] Standard conditions: **1** (0.24 mmol), **2b** (0.20 mmol), Pd(OAc)₂ (10 μ mol), P(3,5-F₂C₆H₃)₃ (18 μ mol), ZnBr₂ (0.20 mmol) in THF at -20 °C for 10 min. [b] Isolated yields.

We next investigated the true structure of the formylated products in the reaction mixture because they may be protected

by the thiolate anion generated *in situ*, leading to the formation of an oxothioacetal **4** (eq 1). In order to obtain structural information on products **3**, we carried out a ^1H NMR experiment of the reaction mixture using **1a** and **2b** under the optimized conditions except for the solvent and reaction time. The ^1H NMR spectra of (i) alkenylzinc reagent **1a**, (ii) thioformate **2b**, (iii) product **3a**, and (iv) the reaction mixture of **1a** and **2b** in THF-d_8 within 5 min are shown in Figure 2. The doublet peak around 10 ppm for the spectrum (iv) was attributed to the unprotected formyl group of **3a**. In addition, we did not observe any peaks around 5–7 ppm, which could be attributed to the signal for oxothioacetal derivatives,¹³ other than that for **1a** in the spectrum (iv). These results indicated that the desired product **3a** had an unprotected formyl group, which can be further utilized for the C–C bond formation in a one-pot system.

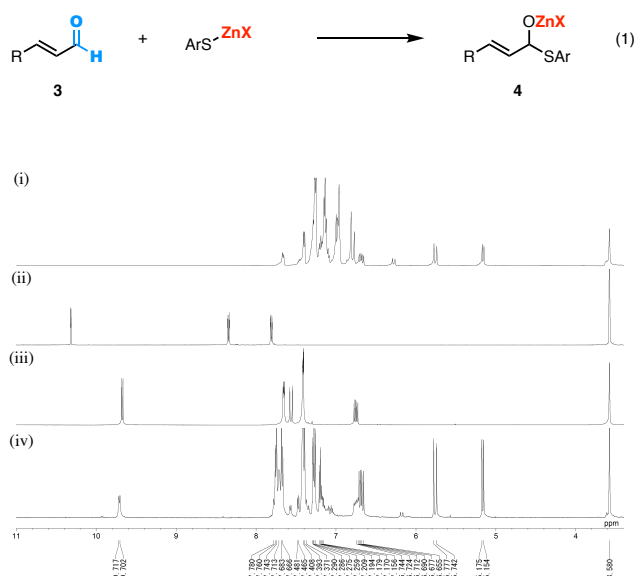
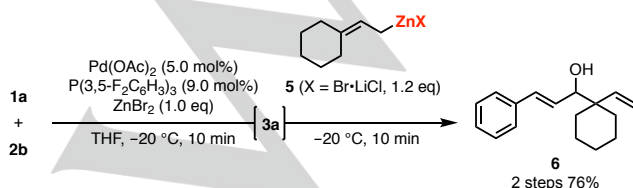


Figure 2. ^1H NMR spectra of (i) **1a**, (ii) **2b**, (iii) **3a**, and (iv) the reaction mixture of **1a** and **2b** in THF-d_8 .

Finally, we present an application of the present reaction, as shown in Scheme 2. After the formylation of **1a** under the optimized reaction conditions, the addition of allylzinc reagent **5** to the reaction mixture with subsequent stirring at $-20\text{ }^\circ\text{C}$ for 10 min afforded 1,5-diene-3-ol **6** in 76% yield over two steps.^[14] This transformation enabled two C–C bond formations in the one-pot reaction, which demonstrated the versatility of the unprotected enal synthesis.



Scheme 2. One-pot transformation: formylation and allylation.

Conclusions

In summary, we have demonstrated the palladium-catalyzed formylation of alkenylzinc reagents using *S*-(4-nitrophenyl) thioformate. Various functional groups were tolerated in the formylation. ^1H NMR analysis of the reaction mixture revealed that the formyl group of the enal product existed in the unprotected form. Thus, the addition of allylzinc reagent **5** to the reaction mixture enabled the subsequent allylation to give the 1,5-diene-3-ol compound. Further investigations including formylation with alkylzinc reagents are underway in our laboratory.

Experimental Section

General procedure for palladium-catalyzed formylation of alkenylzinc reagents **1 with **2**:** The reaction was performed in a 20-mL Schlenk flask equipped with a teflon-coated magnetic stirrer. ZnBr_2 (45 mg, 0.20 mmol) was heated to $140\text{ }^\circ\text{C}$ under high vacuum for 1 h. After cooling to room temperature, $\text{Pd}(\text{OAc})_2$ (2.3 mg, 0.010 mmol) and tris(3,5-difluorophenyl)phosphine (6.7 mg, 0.018 mmol) were added to the vessel. Varying amounts of THF were added in order to reach a total volume of 1.5 mL, and the mixture was stirred at $-20\text{ }^\circ\text{C}$ for 5 min. After the addition of *S*-(4-nitrophenyl) thioformate (**2b**, 37 mg, 0.20 mmol) at $-20\text{ }^\circ\text{C}$, alkenylzinc reagent **1** (0.24 mmol, in THF) was then added at $-20\text{ }^\circ\text{C}$. The resulting mixture was stirred at $-20\text{ }^\circ\text{C}$ for 10 min. The reaction was quenched with saturated aqueous NH_4Cl solution (3.0 mL). The resulting mixture was extracted with ethyl acetate. The combined organic layers were washed with brine, dried over sodium sulfate, and concentrated *in vacuo*. Purification by flash silica gel column chromatography using hexane/ethyl acetate as an eluent gave the corresponding enal **3**.

Keywords: Palladium • Formylation • Organozinc • Cross coupling • Synthetic methods

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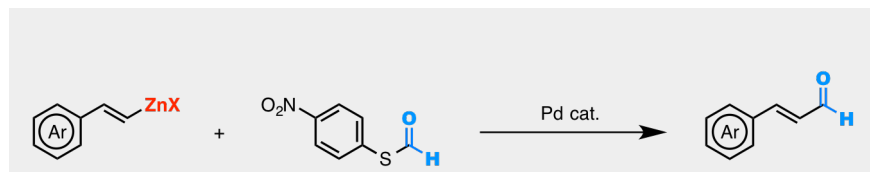
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COMMUNICATION

Key Topic*

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**Palladium-Catalyzed Formylation of
Alkenylzinc Reagents with S-(4-
Nitrophenyl) Thioformate**



Efficient synthesis of enal compounds was accomplished through formylation of alkenylzinc reagents with S-(4-nitrophenyl) thioformate. This reaction proceeded to completion under mild reaction conditions within 10 min, affording various enal derivatives in good-to-high yields

*Formylation